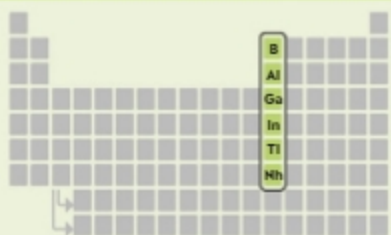


The Boron Group



B
Al
Ga
In
Tl
Nh

Although none of these elements are very reactive, they do differ in other ways. Aluminum is hard and has many uses, including in construction; gallium melts at just 86°F (30°C); and thallium is poisonous.

Pellets of pure aluminum refined in a laboratory



The Carbon Group



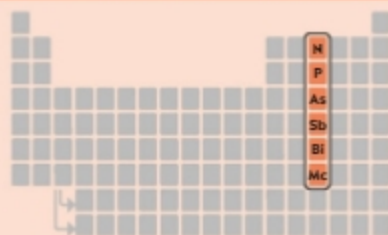
C
Si
Ge
Sn
Pb
Fl



Laboratory sample of pure tin

Group 14 includes the non-metal carbon, the semi-metals silicon and germanium, the metals tin and lead, and unstable, radioactive flerovium—the newest member of the group.

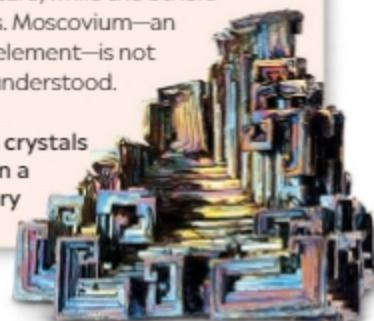
The Nitrogen Group



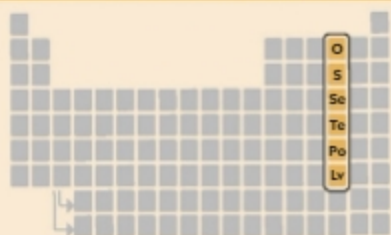
N
P
As
Sb
Bi
Mc

Non-metals, semi-metals, and metals make up the nitrogen group. Nitrogen is a gas at room temperature, while the others are solids. Moscovium—an artificial element—is not yet well understood.

Bismuth crystals refined in a laboratory



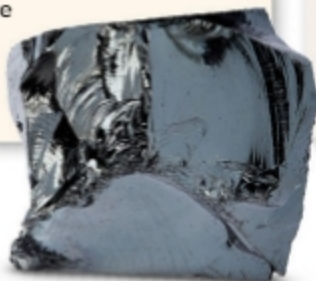
The Oxygen Group



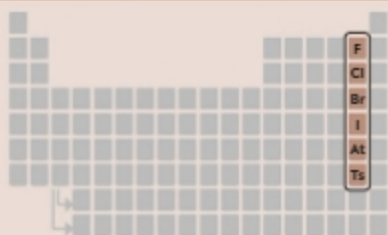
O
S
Se
Te
Po
Lv

The elements in group 16 are diverse. There are two non-metals (oxygen and sulfur), three semi-metals (selenium, tellurium, and polonium), and livermorium, which is a relatively new element.

Chunk of pure selenium refined in a laboratory



Halogens



F
Cl
Br
I
At
Ts

The word *halogen* means “salt forming” because the group 17 elements readily form salts when they react with metals. It contains fluorine, chlorine, bromine, iodine, astatine, and tennessine.

Liquid bromine (bottom) and bromine gas in a glass sphere



Noble Gases



He
Ne
Ar
Kr
Xe
Rn
Og

Colorless, odorless, and tasteless, the group 18 elements are not very reactive. Their reluctance to react with the other elements led to their being called “noble”—the same way that rulers do not interact with their subjects.

Pure argon glowing in response to an electric discharge

